

ultrasound guidance. Patients are evaluated throughout the treatment and post treatment for the occurrence of an infection.

Results: At the participating clinic, a total of 76 APBI patients have successfully completed a full course of treatment. This study was based off of patient evaluations that occurred before and during treatments as well as 1 month and every 6 months after completion of treatment. All patients were started on prophylactic antibiotic since the insertion of the catheter until the completion of the treatment. From the 76 APBI patients a total of 3 showed signs of an infection within the evaluation period. This correlated to 3.9% of all patients at the participating clinic that showed signs of an infection. This is in contrast to previous studies with reported infection rates of 11-14% associated with APBI. These three patients were then evaluated for the timing of the infection. One of the patients developed an infection 79 days post catheter insertion, another patient developed an infection 11 days post catheter insertion, and the final patient developed an infection 289 days post catheter insertion. None of the patients required hospital admission, but they were treated as outpatients on additional antibiotics. Additionally there were a total of 62 CED insertions completed at the participating clinic while there were 14 insertions completed under ultrasound guidance. All of the identified patient's APBI treatments in this study that showed an infection, were performed with the use of a CED. This amounts to 4.8% of the patients with a CED insertion showing signs of an infection. There were no patients with reported infectious complications where ultrasound was used to place the final treatment catheter.

Conclusions: The participating clinic's data shows that 3.9% of patients treated with APBI showed signs of infection. This is significantly lower than previously published data. Based on these infection rates, APBI should still be considered as a viable treatment option for early stage breast cancer patients. The rate of infection in CED based APBI is higher than ultrasound based insertion though the limited number of patients could limit this conclusion.

MISCELLANEOUS SNAP ORAL

Friday, April 10, 2015

4:00 PM - 5:00 PM

MSOR01 Presentation Time: 4:00 PM

I-125 Seeds Continuous Low Dose Rate Radiation Induced Severe DNA Damages on H520 Lung Cancer Cell

Ang Qu, MD¹, Hao Wang, MD¹, Junjie Wang, MD¹, Yuzhu Hou, Ph.D.², Yong Zhao, PhD³. ¹Department of Radiation Oncology, Peking University Third Hospital, Beijing, China; ²Transplantation Biology Research Division, Institute of Zoology, Chinese Academy of Sciences, Beijing, China; ³Transplantation Biology Research Division, Peking University Third Hospital/Institute of Zoology, Chinese Academy of Sciences, Beijing, China.

Purpose: I-125 seeds continuous low dose rate radiation (CLDR) was used to treat patients with lung cancer, but it was not fully understood unlike external beam radiation (EBRT). So we determine the biological effects between EBRT and CLDR in vitro to explore the mechanisms.

Materials and Methods: H520 lung cancer cells were exposed to the same doses by EBRT and CLDR. 6-MV X-ray linear accelerator was used for EBRT and I-125 seeds were used for CLDR. The responses to different radiation were detected by cell survival, death, cycle and apoptosis. Then Molecules related to DSB damage and repair were determined by real-time PCR and western blotting.

Results: H520 cells' cloning efficiency descended rapidly as the radiation dose raised, especially in CLDR group. RBE (CLDR/EBRT) = 1.66. The results were the same as cell growth assay by MTS. 48 hrs after radiation at 4Gy, the percentage of cells in G2/M phase at a higher level in CLDR group by FCM. Additionally, we performed the cell apoptosis assay by FCM with PI/Annexin-V staining. We found markedly increased apoptotic cells upon CLDR. At molecular level, P-H2AX (Ser139) was significantly elevated after CLDR. Either CLDR or EBRT showed little effects on DNA-PK, Ku80, and Ku70 expression at the mRNA level. Furthermore, we found that Ku70 and Ku80 were obviously reduced at

protein levels. The phosphorylations of DNA-PKcs at different sites were found no increased correspondingly.

Conclusions: I-125 seeds CLDR was more efficient to inhibit H520 lung cancer cell growth than EBRT. CLDR led to severe DNA damage by activating H2AX, and reduced DSBs repair through downregulated ku70 and ku80 protein and inhibiting phosphorylated T2609 and S2056 sites in DNA-PKcs on H520 lung cancer cells.

MSOR02 Presentation Time: 4:04 PM

External Beam Radiotherapy Versus Radioactive Implant for Choroidal Melanoma, A Seer (Surveillance Epidemiology and End Results) Analysis

Matthew J. Abrams, MD, Nolan L. Gagne, PhD, Christopher S. Melhus, PhD, John E. Mignano, MD, PhD. Department of Radiation Oncology, Tufts University School of Medicine, Boston, MA, USA.

Purpose: Uveal melanoma is the most common primary intra-ocular malignancy in adults with 4.3 new cases per million people in the U.S., and the most common subtype arises from the choroid. Historically, enucleation was the standard of care for choroidal melanoma until the COMS trial in the 1980's - 90's which showed no difference in survival between appropriately selected patients treated with enucleation or plaque brachytherapy. Since these publications, plaque brachytherapy, and more recently, external beam (including particle therapy) have been used to treat non-metastatic choroidal melanoma. Since there are no large, modern randomized trials comparing these therapies, the optimal treatment modality is unknown.

Materials and Methods: We compared the efficacy of EBRT (external beam radiation therapy, e.g., proton beam, gamma knife radiosurgery, etc.) versus RI (radioactive implant, e.g., plaque brachytherapy, etc.) by searching the Surveillance Epidemiology and End Results (SEER) database for men and women aged 20-79 diagnosed with choroidal melanoma (no microscopic confirmation necessary) from 2004-2011, treated with EBRT or RI. Included patients were clinically T1-T4, N0, M0 (AJCC 6th edition, 2004+). Patients who received other types of radiation or more than one radiation modality were excluded. Patients who received surgery as upfront therapy were also excluded. Overall survival (OS) and cause specific survival (CSS) curves were calculated by the Kaplan-Meier method and univariate and multivariate analyses on covariates were calculated by the log-rank test and Cox proportional hazards ratio model. Chi-squared analysis was performed on categorical variables to test their interaction with radiation technique.

Results: A total of 1001 cases (379 EBRT and 622 RI) in the CSS analysis and 1004 cases (380 EBRT and 624 RI) in the OS were analyzed. Most patients had T2 tumors, received RI and were of white race; there was a nearly even split between men and women. In the OS analysis, there were 41 deaths in the EBRT cohort and 65 deaths in the RI cohort. Using the Kaplan-Meier method and the log-rank test, we found no difference between CSS (5 year CSS 88.3% EBRT vs. 88.3% RI $p = 0.920$) and OS (5 year OS 83.3% EBRT vs. 82.5% RI $p = 0.688$). Examining each tumor stage subgroup individually (T1-T4) we found no statistically significant difference in OS or CSS in each group, however, there was a trend towards improved OS with EBRT for T1 tumors (5 yr OS 94.9% EBRT vs. 86.6% RI $p = 0.08$). We analyzed each covariate (radiation type, sex, histology, tumor stage, age at diagnosis, race and year of diagnosis) and found that tumor stage ($p=0.001$) and age at diagnosis ($p = 0.001$) were significant on univariate analysis by the log rank test. Using the Cox proportional hazards ratio model, we analyzed these significant covariates and found that both age ($p<0.001$) HR(Hazard Ratio)=1.044 [1.026-1.062] and tumor stage ($p=0.001$) T2 HR=1.648 [0.989-2.745], T3 HR 2.845 [1.462-5.534], T4 HR 5.739 [2.141-15.382] remained statistically significant on multivariate analysis.

Conclusions: Tumor stage and age appear to be independent predictors for risk of death. We could not identify a difference in overall survival or cause specific survival between patients treated with EBRT and RI and either modality may be appropriate after full consideration of the inherent treatment toxicities, costs and convenience to the patient.